



5 VOLT DC SUPPLY

FINAL PROJECT PRESENTATION

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OBJECTIVES:-



***DESIGNING
AND
ANALYZING A
LOGICAL
CIRCUIT***



APPARATUS



THEORY

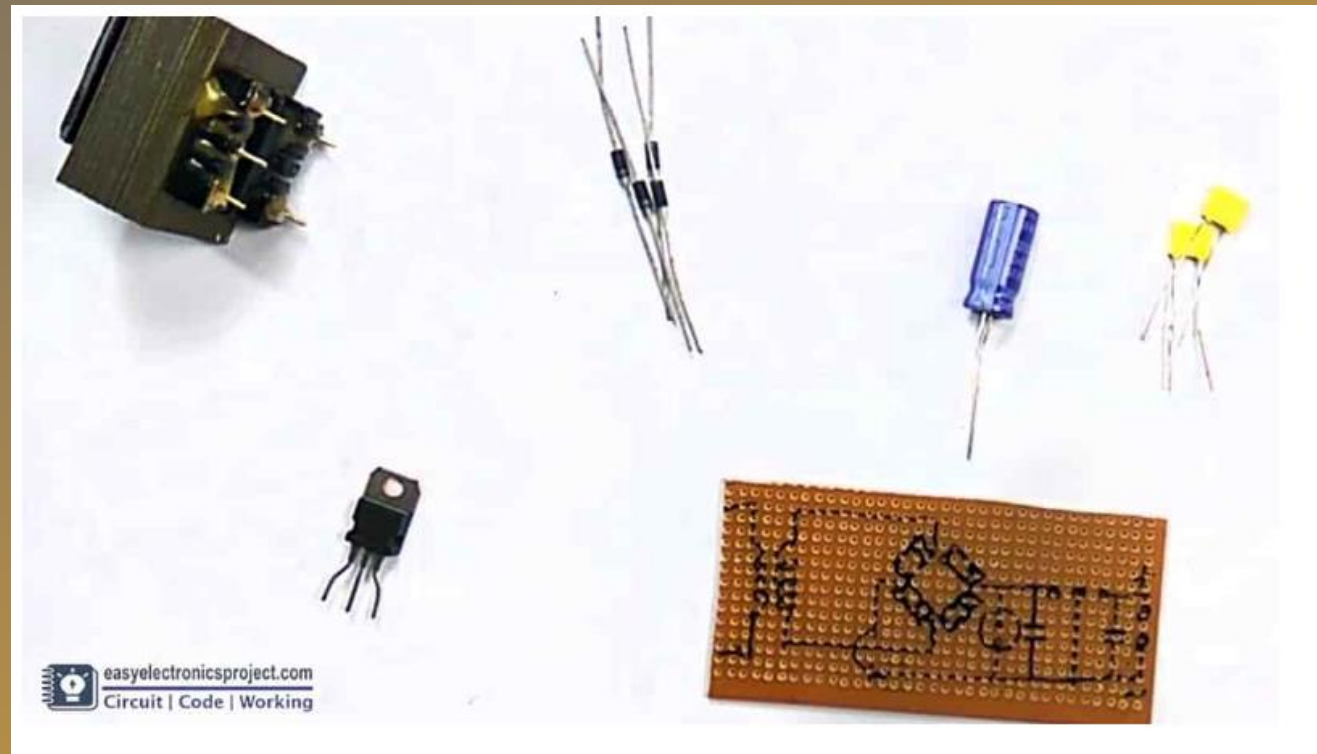
INTRODUCTION:-

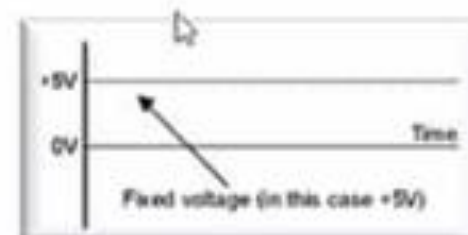
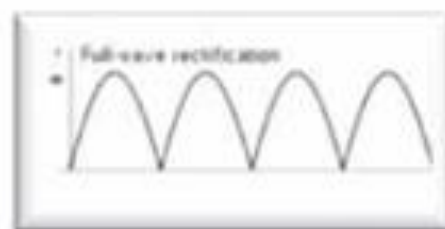
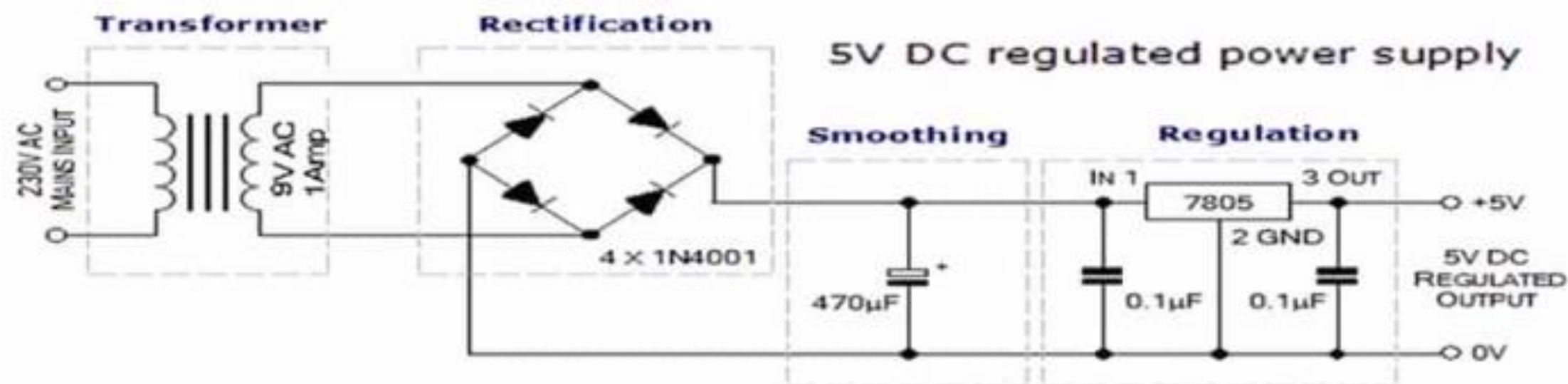
- *5 Volt DC power supplies are one of the most common power supplies in use today. In general, a 5VDC output is obtained from a 220VAC input using a combination of transformers, diodes and transistors. 5V power supplies can be of two types: 5V regulated power supplies, and 5V unregulated power supplies.*
- *Power electronics can be described as the process of applying solid state electronics to control and convert electric power from one form to the other. Power conversion systems under power electronics can be classified according to the type of input and output power. We have:*
 - *AC to DC (known as Rectifier)*
- *All power supplies must have a power input, which receives electrical energy from the energy source and a power output that delivers energy to the load. For this practical, I am considering the AC to DC conversion (Rectification). Using this conversion, the construction of 5-volt regulated power supply which supplies regulated 5volts electric energy to an electric load was carried out*

APPARATUS:-

- *Power supply (220 Volt)*
- *BREAD BOARD*
- *Step down Transformer (12 Volt)*
- *Bridge Rectifier (4 diodes)*
- *Capacitor Filter (Capacitor 470nF)*
- *Voltage Regulator Integrated Circuit (LM7805)*
- *Load Resistance (1k)*
- *Small LED()*

APPARATUS IMAGES:-





Operation of the Circuit...



easyelectronicsproject.com
Circuit | Code | Working

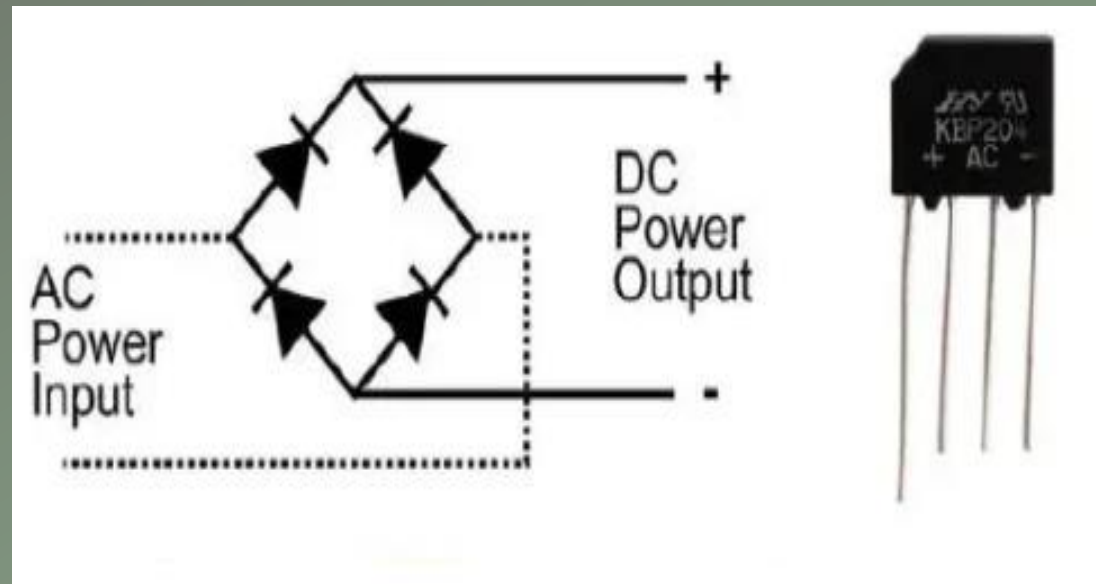
Step down Transformer (12 Volt):

The step down transformer is used transform the incoming line voltage down to the required level for the power supply. It steps down the 220 voltage from the main supply isolates the output circuit from the line supply. The current rating and the secondary rating is an important factor, which depends upon the current needed by the load. So I decided to choose a 15 volt secondary voltage transformer with current rating of 2A.



Bridge Rectifier (4 diodes):

The rectifier converts the incoming AC Electrical signals to DC signals. The bridge rectification is done by soldering four 1N 4007 diodes as they have the ability of with standing a higher feedback voltage. It has a current rating of 1A and a peak reverse voltage of 50v.



Capacitive Filter (470nF):

The DC from the rectifier is not constant, the waveform crosses the zero axis. For this reason, a capacitor is used to fill in the troughs in the waveform. It removes the ripples and smoothens the signals, enabling the next stage of the power supply to operate.



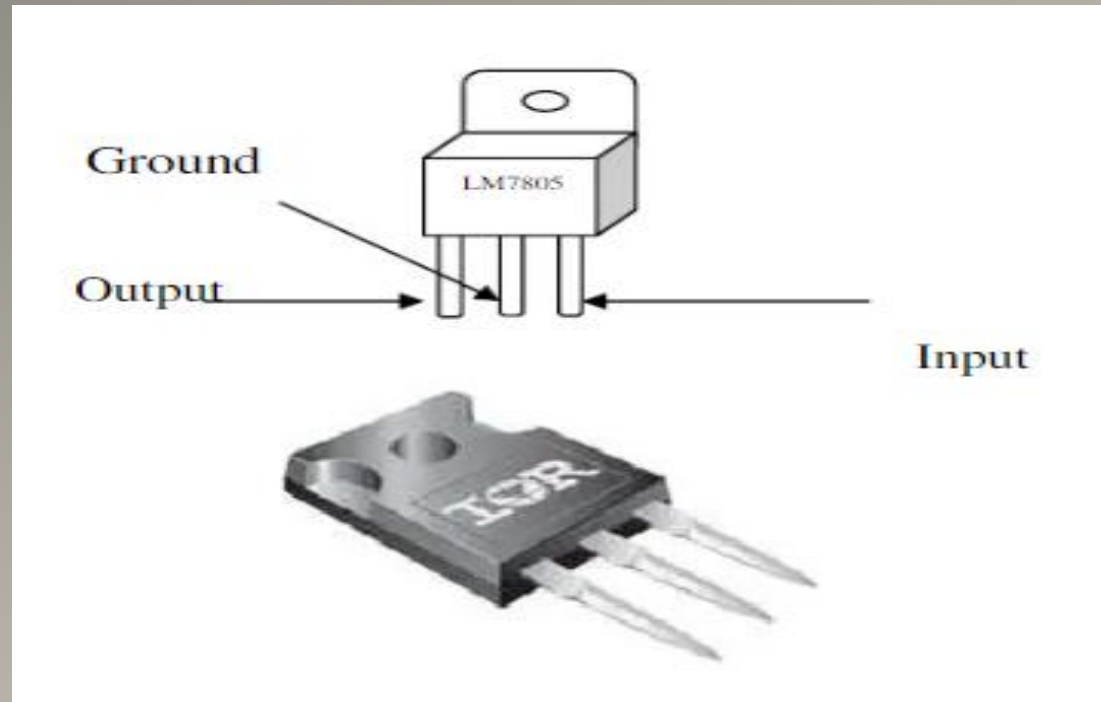
$f = \text{frequency of AC (50Hz)}$



Voltage Regulator (LM7805):

This stage of power supply takes the smoothed voltage from the capacitive filter and uses a regulator to produce a constant output regardless of any fluctuations at the input level. This particular IC has the following characteristics which makes it suitable for use

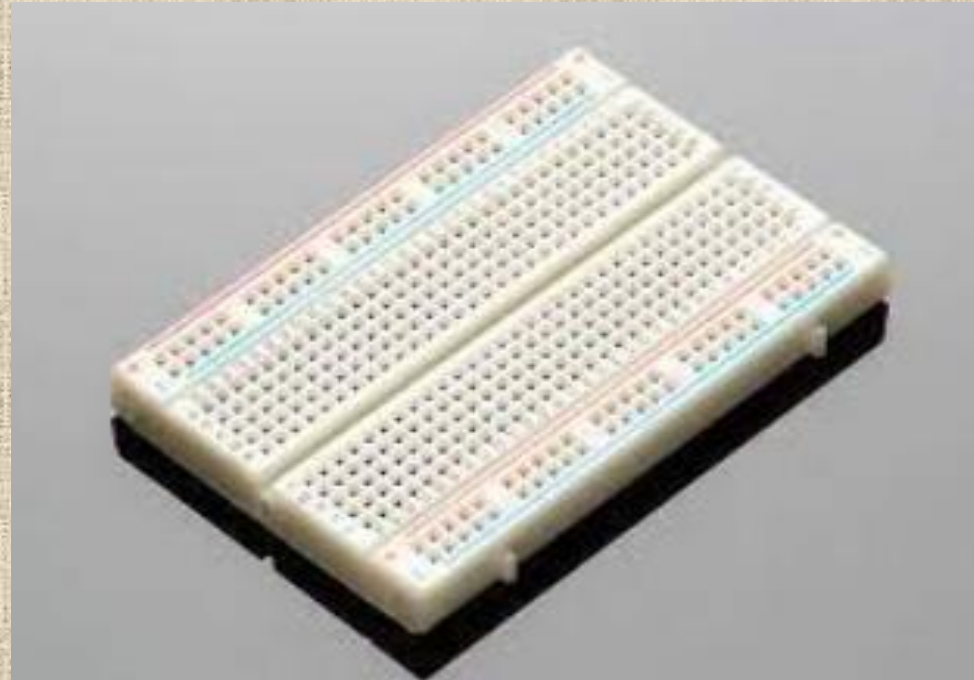
- Output voltage range $V(\text{Max}) = 5.2\text{v}$; $V(\text{Min}) = 4.8\text{v}$
- Input voltage range $7\text{ V} - 35\text{ V}$
- Current rating $I = 1\text{A}$



BREAD BOARD:

This is a board on which the circuit elements are arranged. There are two major types;

- Bread board (which is used for temporary circuit)*
- Vero board (which is used for permanent circuit) For this Circuit, a Vero Board was recommended for easy soldering of components. But we use here a bread board.*



LOAD RESISTANCE & LED:

A base load element is an electrical component which provides a certain minimum load. In many lighting installations, dimmers and electronic transformers are used, which require a certain minimum load to operate correctly. This is not always achieved by the capacitive power supplies of the connected LED lamps.



WORKING OF THE PROJECT:-

4 Steps to Convert 230V AC to 5V DC:

■ 1. Step Down the Voltage Level:

The step-down converters are used for converting the high voltage into low voltage. The converter with output voltage less than the input voltage is called as a step-down converter, and the converter with output voltage greater than the input voltage is called as step-up converter. There are step-up and step-down transformers which are used to step up or step down the voltage levels. 230V AC is converted into 12V AC using a step-down transformer. 12V output of stepdown transformer is an RMS value and its peak value is given by the product of square root of two with RMS value, which is approximately 17V.

2. Convert AC to DC:

- *230V AC power is converted into 12V AC (12V RMS value wherein the peak value is around 17V), but the required power is 5V DC; for this purpose, 17V AC power must be primarily converted into DC power then it can be stepped down to the 5V DC. AC power can be converted into DC using one of the power electronic converters called as Rectifier. There are different types of rectifiers, such as half-wave rectifier, full-wave rectifier and bridge rectifier. We are using a full wave rectifier for converting AC to DC. Full wave rectifier consists of 2 diodes. We know that the diode is an uncontrolled rectifier which will conduct only forward bias and will not conduct during the reverse bias. If the diode anode voltage is greater than the cathode voltage then the diode is said to be in forward bias. During positive half cycle, diodes conduct and during negative half cycle diodes will not conduct. Thus, AC is converted into DC; here the obtained is not a pure DC as it consists of pulses. Hence, it is called as pulsating DC power. But voltage drop across the diodes is 0.7V.*

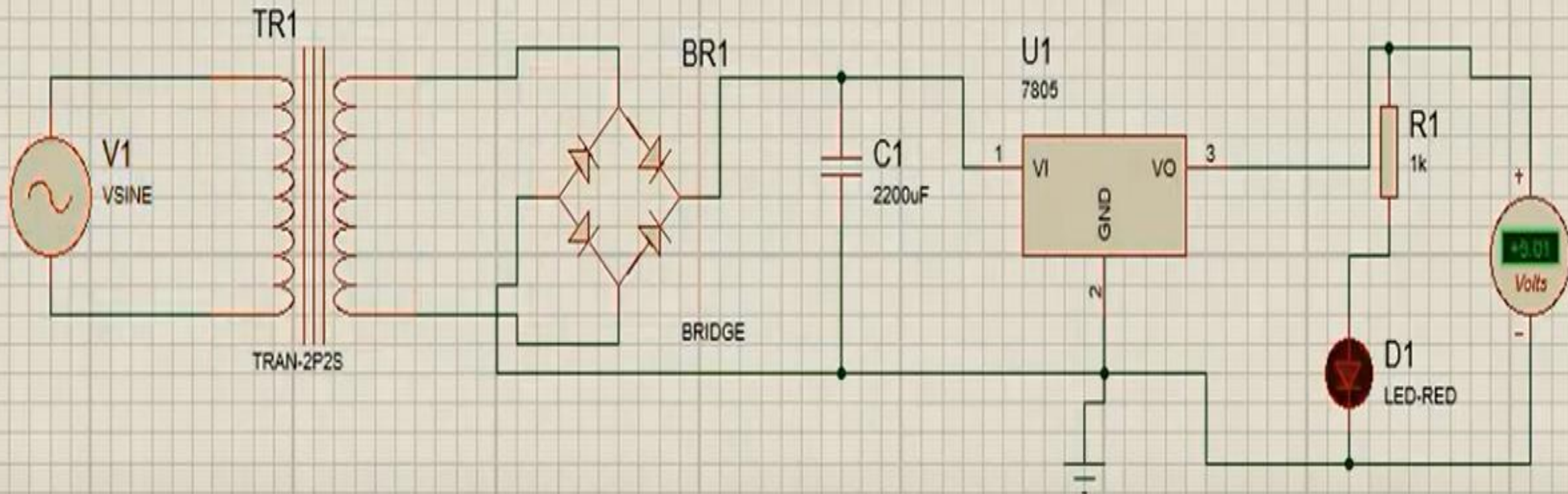
3. Smoothing the Ripples using Filter:

- *15V DC can be regulated into 5V DC using a step-down converter, but before this, it is required to obtain pure DC power. The output of the diode bridge is a DC consisting of ripples also called as pulsating DC. This pulsating DC can be filtered using an inductor filter or a capacitor filter or a resistor-capacitor-coupled filter for removing the ripples. Consider a capacitor filter which is frequently used in most cases for smoothing. We know that a capacitor is an energy storing element. In the circuit, capacitor stores energy while the input increases from zero to a peak value and, while the supply voltage decreases from peak value to zero, capacitor starts discharging. This charging and discharging of the capacitor will make the pulsating DC into pure DC.*

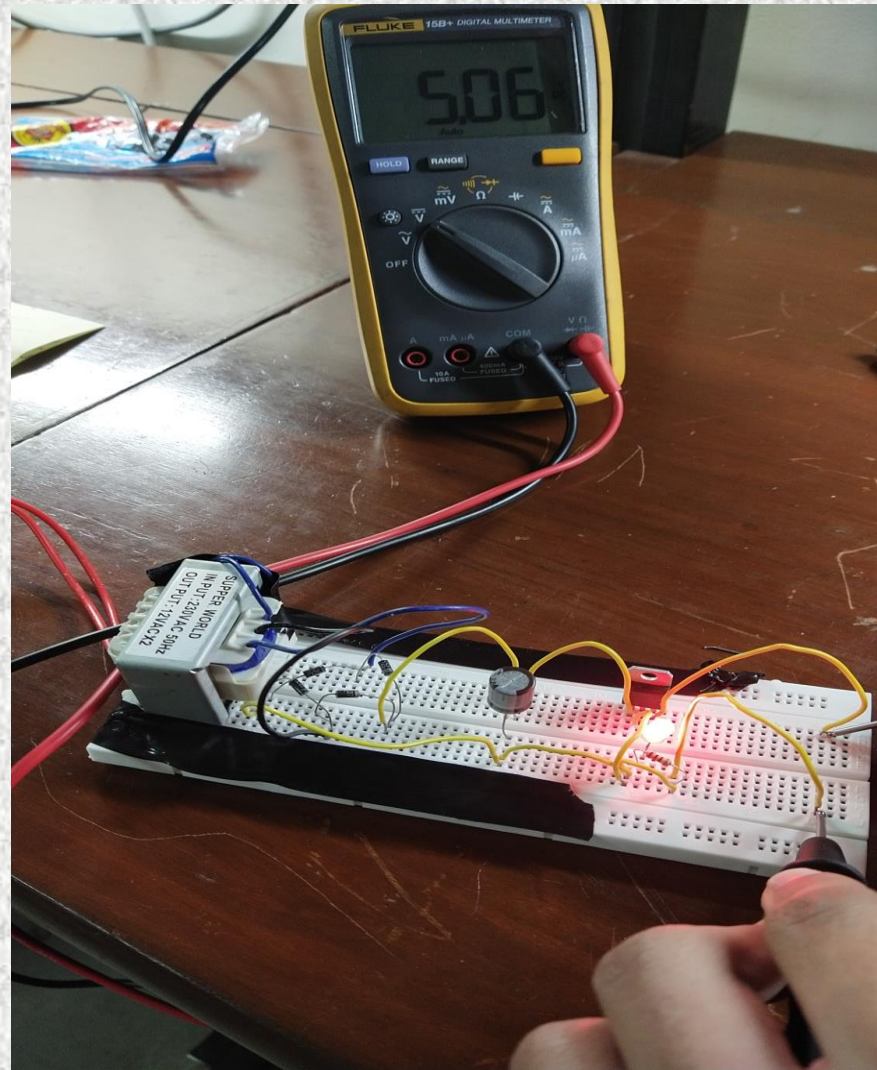
4. Regulating 12V DC into 5V DC using Voltage Regulator:

- *15V DC voltage can be stepped down to 5V DC voltage using a DC step-down converter called as voltage regulator IC7805. The first two digits '78' of IC7805 voltage regulator represent positive series voltage regulators and the last two digits '05' represents the output voltage of the voltage regulator. In general, an IC7805 regulator can withstand voltage ranging from 7.2V to 35V and gives maximum efficiency of 7.2V voltage and if the voltage exceeds 7.2V, then there is loss of energy in the form of heat. To protect the regulator from over heat, thermal protection is provided using a heat sink. Thus, a 5V DC is obtained from 230V AC power.*

PROTEUS IMAGE:-



HARDWARE IMAGE:-



RESULTS:-

When checking with the DMM(Digital Multimeter), the voltage across the load resistance is almost near the 5 volt but the DMM shows 5.06V.

Conclusion:-

- *The power supply circuit was built with a step-down transformer, which steps down the voltage from 230V to 12V AC. The AC voltage was rectified to DC using the bridge rectifier. The capacitors remove the ripples and then voltage was regulated to +5V by the IC LM7805.*